Taewoong Yoon

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EDUCATION

Seoul National University (SNU), Seoul, Korea
MS/Ph.D Candidate in Physics, Mar 2019 - Present
Thesis Advisor: Prof. Hyunyong Choi

Pohang University of Science and Technology (POSTECH), Pohang, Korea
Bachelor of Science (BS) in Physics, Mar 2014 - Feb 2019

Seoul Science High School (SSHS), Seoul, Korea
Mar 2011 - Feb 2014

RESEARCH INTERESTS

Quantum Optics, Quantum Sensing, Quantum Simulation, Diamond NV Center

RESEARCH EXPERIENCE

Ultrafast Quantum Photonics Laboratory, SNU
Graduate Research Assistant (Advisor: Prof. Hyunyong Choi) Aug 2019 - Present

- Constructed room- and low-temperature confocal setups for NV center experiments
- Conducted defect generation research using femtosecond laser single pulse
- Implemented NV axis discrimination method using microwave polarization
- Performed quantum simulation studies on NV and P1 dipolar spin systems

Quantum-Field Laser Laboratory, SNU
Undergraduate Research Intern (Advisor: Prof. Kyungwon An) Jun 2018 - Aug 2018

- Assisted with the acoustic Penrose cavity experiment
- Optimized simulated stadium cavity eigenmodes data using Python

Quantum Optics & Quantum Information Group, POSTECH
Undergraduate Research Intern (Advisor: Prof. Yoonho Kim) Sep 2016 - Aug 2017

- Built a compact Michelson interferometer via 3D printing

Center for Quantum Information, KIST
Research Intern Jun 2016 - Aug 2016

- Studied nonlinear optics and quantum information theory

PUBLICATIONS

1. **Identifying NV center axes via spatially varying microwave fields for vector magnetometry**, Taewoong Yoon, Myungjun Cha, Dohun Kim, and Hyunyong Choi, *Applied Physics Letters*, **126**, 144002 (2025). https://doi.org/10.1063/5.0243162

CONFERENCES & PRESENTATIONS

Oral Presentations

1. Probing and controlling many-body dipolar interactions between electron spins in diamond, *KPS Spring Meeting*, Daejeon, Korea, Apr 2024.
2. Precise calibration of magnetic field vector in a diamond nitrogen-vacancy center ensemble, *Optics and Photonics Congress (OPC)*, Jeju, Korea, Aug 2023.
3. Aberration correction of femtosecond lasers for deterministic quantum emitters in nitrogen-vacancy diamonds, *Optics and Photonics Congress (OPC)*, Jeju, Korea, Jul 2021.

Poster Presentations

1. Deterministic creation of single nitrogen-vacancy center in diamond using femtosecond laser writing, *Conference on Lasers and Electro-Optics (CLEO)*, San Jose, USA, May 2022.

HONORS & AWARDS

1. **BK Excellent TA award,** Jul 2020
2. **BK Frontier Fellowship**, Mar 2019 - Aug 2019
3. **The National Scholarship for Science and Engineering**, Mar 2014 - Feb 2018

EXTRA CURRICULAR ACTIVITIES

Teaching assistant at SNU

- Electrodynamics I, 2021 Spring
- Computational Physics, 2020 Fall
- Electromagnetic Waves and Optics, 2020 Spring
- Physics Lab. 2, 2019 Summer, Fall
- Physics Lab. 1, 2019 Spring

Student Mentoring Program (SMP) at POSTECH

- General Physics II, 2017 Fall
- General Physics I, 2017 Spring